Detailed AWS Cost Calculation — 24×7 Deployment

1. Assumptions

For this calculation, the HRMS application is assumed to run continuously:

```
24 hours/day
× 30 days/month
= 720 hours/month
```

Proposed AWS configuration

Service	Configuration
EC2	1 × `t3.small`, Linux
RDS PostgreSQL	`db.t3.micro`, Single-AZ
RDS Storage	20 GB
EBS	20 GB gp3
ECR	10 GB Docker images
Secrets Manager	4 secrets
S3	20 GB files/documents
CloudWatch	Application/system logs + basic monitoring
SES	Low-volume HRMS emails
SNS	Low-volume notifications
VPC	VPC, subnets, route tables, security groups
ACM	Public SSL certificate
Fargate	Not used

> **Important:** AWS pricing varies by region and usage. The calculations below are planning estimates, not a final AWS invoice. AWS also states that RDS pricing depends on region, instance type, deployment option, storage, backups, I/O, and other factors.

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2. EC2 — Application Server

Configuration

```
Instance: t3.small
OS: Linux
Instances: 1
Running: 24×7
```

AWS lists the `t3.small` at **$0.0209/hour** in its US East (N. Virginia) example pricing. The actual Hyderabad price can differ.

Calculation

```
24 hours × 30 days
= 720 hours/month

$0.0209 × 720
= $15.048/month
```

EC2 estimated compute cost

≈ $15.05/month

EC2 may also have additional charges depending on public IPv4 usage, data transfer, CPU-credit usage under T3 Unlimited, etc.

Planning value

≈ $15–20/month

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3. RDS PostgreSQL — Database Server

Configuration

```
Engine: PostgreSQL
Instance: db.t3.micro
Deployment: Single-AZ
Storage: 20 GB
Running: 24×7
```

RDS On-Demand DB instances are billed according to the time the DB instance runs; AWS supports per-second billing with a minimum billable duration after status changes.

Instance-hours

```
24 hours × 30 days
= 720 hours/month
```

If the Hyderabad `db.t3.micro` rate were represented as **R dollars/hour**, then:

```
RDS Compute Cost
= R × 720
```

For example, if the actual rate were $0.02/hour:

```
$0.02 × 720
= $14.40/month
```

RDS storage

```
20 GB
× RDS storage price per GB-month
= Monthly storage cost
```

Complete RDS calculation

```
RDS Total
=
DB Instance Cost
+ 20 GB Storage
+ Backup Storage
+ I/O
+ Data Transfer, if applicable
+ CPU Credits, if applicable
```

For T3 RDS instances, AWS notes that CPU-credit charges can apply when average CPU utilization exceeds the baseline under Unlimited mode.

Planning value

For a small HRMS workload, a reasonable initial planning range is:

≈ $15–25/month

This should be verified in the AWS Pricing Calculator for **Asia Pacific (Hyderabad)** before deployment.

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4. EBS — EC2 Disk Storage

Configuration

```
Volume: gp3
Size: 20 GB
```

AWS charges EBS based on the amount of storage provisioned per month. gp3 includes baseline performance of 3,000 IOPS and 125 MB/s throughput; additional IOPS/throughput can cost extra.

AWS gives an example gp3 price of **$0.08/GB-month** in a region using that rate.

Calculation

```
20 GB × $0.08
= $1.60/month
```

EBS estimated cost

≈ $1.60/month

If snapshots are added, snapshot storage can increase the cost.

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5. ECR — Docker Image Storage

Configuration

```
Docker image storage: 10 GB
```

ECR is primarily charged for the amount of container image data stored.

Planning calculation:

```
10 GB × ECR storage rate
= Monthly ECR storage cost
```

For example, using an illustrative rate of $0.10/GB-month:

```
10 GB × $0.10
= $1.00/month
```

ECR planning value

≈ $1–2/month

The actual cost depends on the storage and applicable regional pricing.

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6. Secrets Manager

Configuration

```
Number of secrets: 4
Average duration: 30 days
```

Example secrets:

```
1. Database username/password
2. JWT/application secret
3. AWS/S3 application credentials or configuration
4. Other application secret
```

Calculation:

```
4 secrets
× Secrets Manager price per secret/month
+ API usage
```

If the planning rate is approximately $0.40/secret/month:

```
4 × $0.40
= $1.60/month
```

Secrets Manager planning value

≈ $1.60/month + API usage

Your previous AWS estimate used 8 secrets and showed **$3.20/month**, which corresponds to $0.40 per secret/month in that estimate.

Therefore, reducing from 8 secrets to 4 would approximately give:

```
4 × $0.40
= $1.60/month
```

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7. S3 — File and Document Storage

Configuration

```
Storage: 20 GB
Storage Class: S3 Standard
```

S3 pricing consists of several components, including storage, requests, data transfer, and other features.

For a simple planning calculation, AWS documentation gives an example S3 Standard rate of **$0.10/GB-month** for the first tier.

Storage calculation

```
20 GB × $0.10
= $2.00/month
```

Then add:

```
PUT requests
GET requests
Other requests
Data transfer, if applicable
```

S3 planning value

≈ $2–3/month

If the HRMS stores employee documents, salary slips, certificates, profile images, etc., S3 is useful because it provides durable object storage.

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8. CloudWatch — Monitoring and Logs

CloudWatch does not simply charge:

```
720 hours × hourly price
```

Instead, the cost depends on things such as:

```
Log ingestion
+
Log storage
+
Metrics
+
Alarms
+
Dashboards
```

For a small HRMS application, assume:

```
Low-volume application logs
+
Basic metrics
+
Few alarms
```

Planning value

≈ $1–5/month

It could be higher if the application generates large amounts of logs.

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9. SES — Email Service

SES is used for HRMS emails such as:

```
Password reset
Leave approval
Leave rejection
Employee notifications
HR notifications
Account creation
```

SES is usage-based.

Example:

```
1,000 emails/month
× SES price per email
= Monthly email cost
```

For low-volume application emails, the cost can be very small.

Planning value

≈ $0–1/month

The actual cost depends on the number of emails and sending configuration.

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10. SNS — Notification Service

SNS is also usage-based.

Example:

```
10,000 notification/message requests
× Applicable SNS price
= Monthly SNS cost
```

For low-volume HRMS notifications:

Planning value

≈ $0–1/month

The actual amount depends on message volume and delivery type.

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11. VPC

Configuration

```
1 VPC
Subnets
Route tables
Internet Gateway
Security Groups
Network ACLs
```

Basic VPC components themselves generally do not create a significant hourly charge.

Therefore:

```
VPC Basic Infrastructure
≈ $0/month
```

Important

Be careful with additional networking services such as:

```
NAT Gateway
VPN
Transit Gateway
Network Firewall
```

These can introduce additional costs.

Planning value

≈ $0/month for basic VPC components.

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12. ACM — SSL Certificate

AWS Certificate Manager can provide a public SSL/TLS certificate for supported AWS services.

Configuration

```
HTTPS/SSL certificate
```

Planning value

≈ $0/month

ACM itself is therefore not expected to be a significant cost in this architecture.

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13. Fargate

For this architecture:

```
Fargate = Not Used
```

The backend runs directly on EC2.

Therefore:

```
Fargate Cost
= $0/month
```

Your previous AWS calculator estimate also showed Fargate at **$0/month**.

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14. Estimated Monthly Cost — 24×7

AWS Service	Configuration	Estimated Monthly Cost
EC2	1 × `t3.small`, Linux, 720 hrs	**$15–20**
RDS PostgreSQL	`db.t3.micro`, Single-AZ, 20 GB	**$15–25**
EBS	20 GB gp3	**~$1.60**

AWS Service	Configuration	Estimated Monthly Cost
ECR	10 GB images	**$1–2**
Secrets Manager	4 secrets	**~$1.60**
S3	20 GB Standard	**$2–3**
CloudWatch	Logs + monitoring	**$1–5**
SES	Low email volume	**$0–1**
SNS	Low notification volume	**$0–1**
VPC	Basic components	**~$0**
ACM	SSL certificate	**~$0**
Fargate	Not used	**$0**

Estimated total

Using the ranges above:

```
Minimum:
$15 + $15 + $1.60 + $1 + $1.60 + $2 + $1
+ $0 + $0
≈ $37.20/month
```

```
Maximum:
$20 + $25 + $1.60 + $2 + $1.60 + $3 + $5
+ $1 + $1
≈ $60.20/month
```

Expected Monthly AWS Budget

≈ $37–60/month

For planning purposes, it would be safer to keep:

≈ $50–70/month

because actual usage of S3, CloudWatch, SES, SNS, data transfer, backups, and other services can vary.

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15. Approximate Cost in Indian Rupees

Using an example planning exchange rate of:

```
$1 ≈ ₹90
```

Monthly

```
$37 × ₹90
= ₹3,330

$60 × ₹90
= ₹5,400
```

Therefore:

≈ ₹3,300–₹5,400/month

A safer operational budget would be:

≈ ₹5,000–₹7,000/month

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16. Annual Cost

Using the estimated range:

```
$37 × 12
= $444/year

$60 × 12
= $720/year
```

At ₹90/USD:

```
$444 × ₹90
= ₹39,960/year
```

```
$720 × ₹90
= ₹64,800/year
```

Estimated Annual Budget

≈ ₹40,000–₹65,000/year

A safer budget including usage variations would be approximately:

₹60,000–₹85,000/year

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17. Final Architecture Cost Summary

```
                    HRMS APPLICATION
                            │
                            ▼
                    ┌─────────────┐
                    │    EC2      │
                    │  t3.small   │
                    │  24 × 7     │
                    └──────┬──────┘
                            │
             ┌─────────────┼─────────────┐
             ▼              ▼              ▼
        PostgreSQL      S3           CloudWatch
        db.t3.micro   Documents        Logs
             │
             │
      ┌──────┴──────┐
      ▼              ▼
    SES            SNS
   Emails       Notifications

        EC2 also uses:
        ├── EBS
        ├── ECR
        ├── Secrets Manager
        ├── VPC
        └── ACM
```

Final Recommendation

For an initial **24×7 HRMS deployment**, the minimum-cost architecture can be:

```
EC2        → t3.small
RDS        → db.t3.micro, Single-AZ
EBS        → 20 GB gp3
S3         → 20 GB
ECR        → ~10 GB
Secrets    → 4 secrets
CloudWatch → Basic monitoring/logs
SES        → Low-volume emails
SNS        → Low-volume notifications
VPC        → Basic VPC
ACM        → SSL certificate
Fargate    → Not required
```

Expected Cost

Approximately $37–60/month

or approximately

₹3,300–₹5,400/month

with a recommended budget of approximately **₹5,000–₹7,000/month** for initial planning.

Note: Your uploaded AWS Pricing Calculator estimate was much higher at **$9,955.95/month**, primarily because it used an `RDS db.m5.12xlarge` with **Multi-AZ** and 100% utilization, producing **$9,925/month for RDS alone**. The optimized configuration above removes that oversized database configuration and uses a much smaller Single-AZ database.